Dollar as a Safe Asset

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Contents

Dollar Dominance

Safe haven status:

- US financial markets are deep and liquid
- Global reserve currency
- US government and Fed viewed as stable, unlikely to default
- Global demand during stress

During crises, observe ***Flight to Safety***:

- **The dollar appreciates/strengthens during recessions, crises, market stress**
- Risk: investors reallocate portfolios toward safe, liquid assets (USD, US treasuries)
- To buy US treasuries, investors must buy US dollars (treasures are dollar-denominated)
- Surge in demand causes dollar to appreciate

Series

Exchange Rates

- Relative: 1 currency appreciates the other depreciates relative to it.
- Determined by supply & demand.
- BASE/QUOTE
  - Quoted number rises: base strengthens
  - Falls: base weakens
- EUR/USD = $1.10
  - 1 euro = $1.10
  - Inc: EUR appreciating, USD depreciates
- USD/JPY = 150
  - $1 = 150 yen
  - Inc: USD appreciates, JPY depreciates

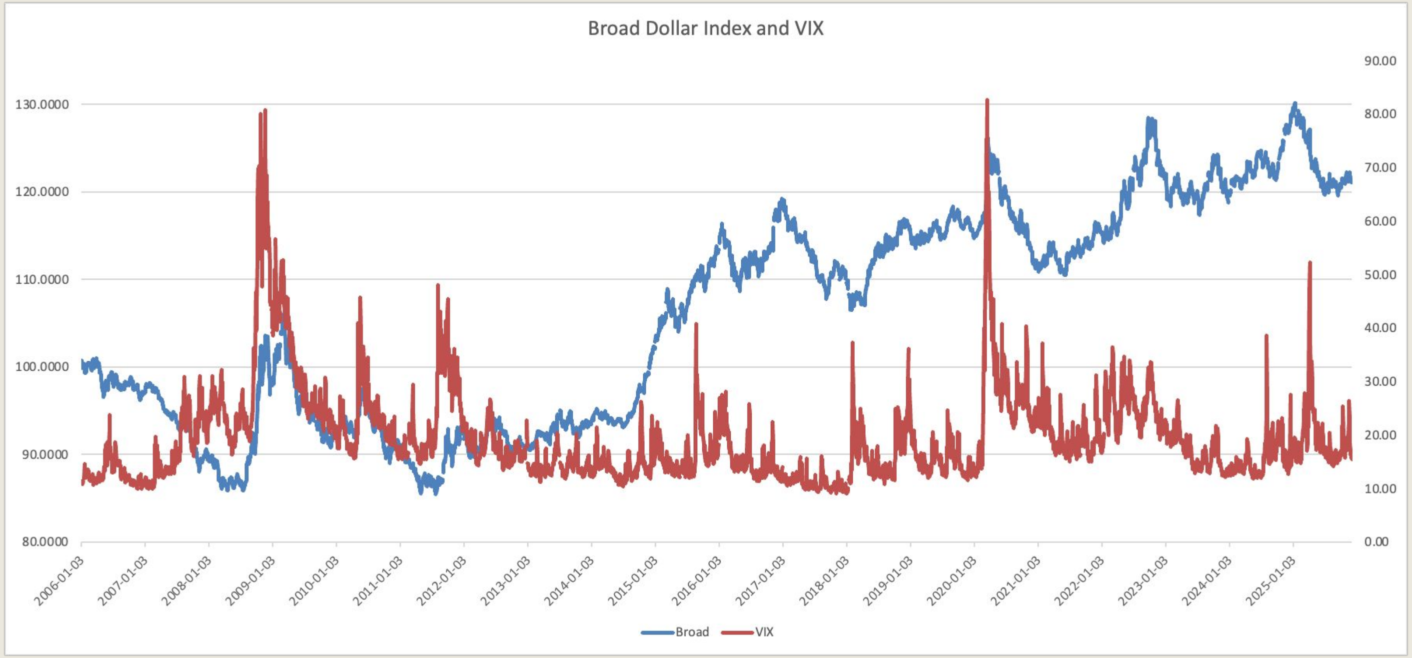
Broad Dollar Index (FED)

- How strong is the dollar compared to the currencies of all the countries the U.S. trades with?
- Weighted average based on each country's trade share.
- ~26 currencies
- Compared against base year Jan 2006 = 100.
  - Above 100: appreciated
  - Below: depreciated
- Rises: Appreciation, during global downturns.
- Falls: Dollar is weakening

VIX

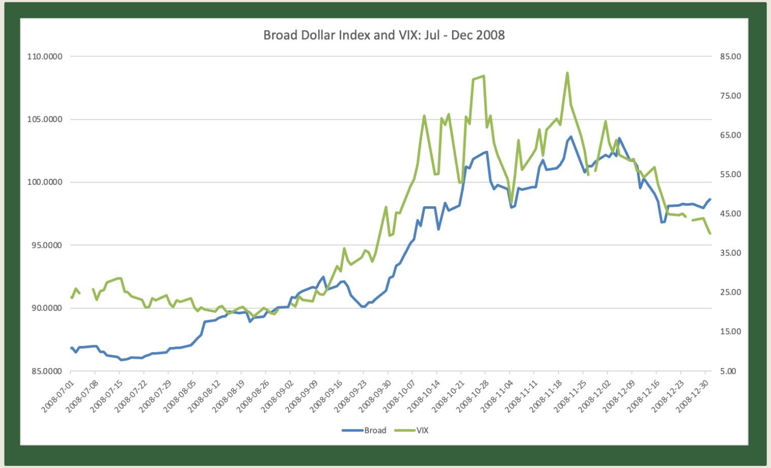
- Measure of the expected volatility of the US stock market over the next 30 days
- Based on S&P 500 index options
- Market's fear gauge
  - Spikes: anxious investors, falling markets
  - Declines: markets are calm & confident
- Levels:
  - < 12: Calm
  - 12 – 20: Normal
  - 20 – 30: Elevated
  - > 30: High fear

Dollar Index and VIX – Long Run

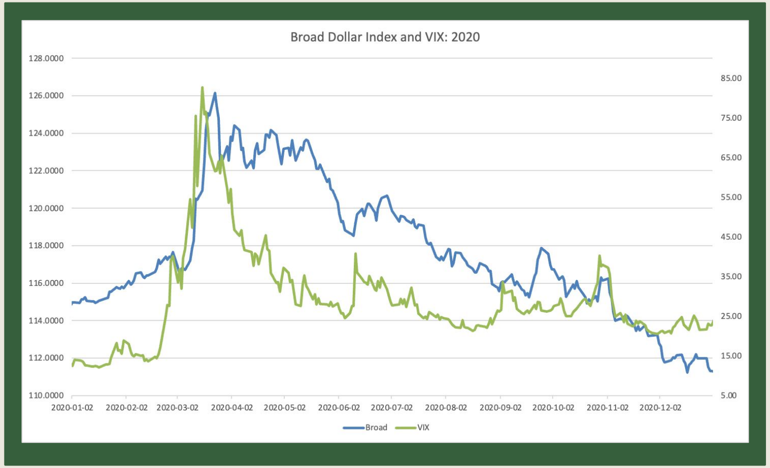


Closer Look:

GFC

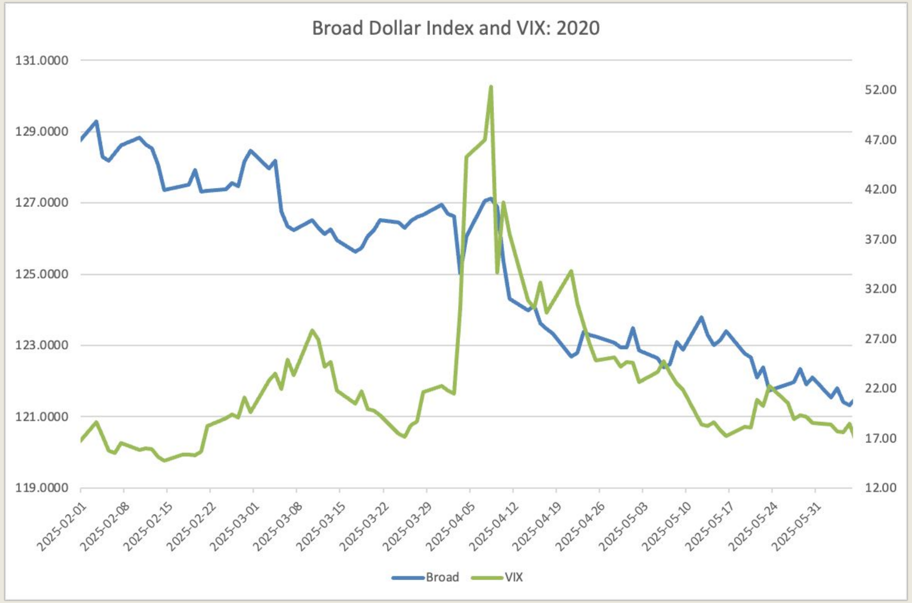


COVID



Dollar appreciation is largely a crisis phenomenon

Liberation Day Tariffs



Flips – dollar depreciation

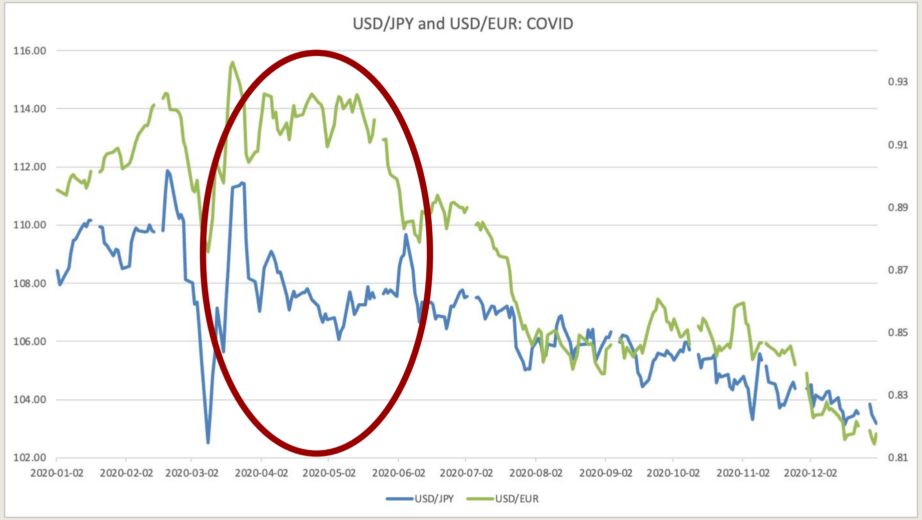
- April 2025: Trump announced steep tariffs, sparking financial market turmoil
- Abnormal: Instead of strengthening, the dollar weakened
- Undermined dollar's traditional role as a safe-haven

What does this shift mean?

- Questions the safety and predictability of US policy
- Perceived weakening of the Fed's independence
- Fractures in alliances with key trading partners
- Increased trade barriers and fiscal deficits

Exchange Rates

- The dollar tends to appreciate during crises because it is viewed as a safe asset
- It also often appreciates against foreign currencies
  - Different responses depending on the currency
  - Risky vs safe currencies
- COVID:
  - USD initially appreciates against EUR and JPY
  - Appreciation against JPY reverted quickly (another a safe-haven currency)



Covered Interest Parity Deviations

Explaining Covered Interest Parity Deviations

What is Covered Interest Parity?:

- Covered Interest Parity is a fundamental principle of no-arbitrage that traditionally held before the 2008 Great Financial Crisis
- The dollar interest rate a borrower pays when directly borrowing in the cash market must equal the implied dollar interest rate obtained in the foreign exchange (FX) swap market
- This principle can be expressed through a multiperiod equation: $(1+y^{\$}_{t,t+n})^n = (1+y^{i}_{t,t+n})^n \frac{S^{i/\$}_t}{F^{i/\$}_{t,t+n}}$
    - $y^{\$}_{t,t+n}$ represents the yield on direct USD borrowing
    - $y^{i}_{t,t+n}$ represents the yield charged to foreign borrowers in their local currency
    - $S^{i/\$}_t$ represents the spot rate at which a foreign borrower swaps his currency for USD
    - $F^{i/\$}_{t,t+n}$ represents the forward rate at which the borrower swaps USD back for his local currency

Before 2008, CIP held because any fluctuation between the direct rate and the implied borrowing rate would be exploited through arbitrage:

- For example, if the implied rate in the EUR/USD swap market was higher than the direct rate, arbitrageurs would directly borrow USD and then lend it in the swap market for Euros.
- In terms of supply and demand, as more arbitrageurs lend USD in the swap market, the price (or yield) of USD in the offshore market for foreign borrowers decreases until an equilibrium is reached at the lower bound of the cash rate
- This worked consistently since lending in the swap market was considered riskless and not subjected to any regulatory constraints

Explaining Covered Interest Parity Deviations

However in the wake of the 2008 Great Financial Crisis, CIP broke down:

- One major reason for this break down of CIP is the institution of restrictions on offshore dollar funding by the Basel Committee on Banking Supervision in its new package of financial regulations (BASEL III)
    - Prior to BASEL III, capital regulations targeting major banks mainly focused on risk-weighted assets and ensuring that sufficient capital was available to hedge
    - Short-term swap or CIP-arbitrage positions were considered essentially riskless and thus exempt.
    - However, BASEL III required that banks maintain collateral against all assets, regardless of risk characteristics in order to maintain a leverage ratio much higher than previous levels
- Essentially, the ability of banks to supply and lend dollars was capped significantly, changing the supply of offshore USD funding from inelastic to elastic

CIP deviations sustained post-2008 due to sustained demand for USD funding by foreign borrowers

- One key demographic for USD swap funding is investors in low interest rate countries like Switzerland and Japan looking to invest in US treasury bonds
    - These investors seek to reduce FX risk by engaging in swap contracts in order to lock in their USD yields at an agreed-upon forward rate
- Additionally, foreign banks tend to seek USD funding in the swap market during times of crisis (USD funding from prime MMFs tends to dry up) in order to continue financing dollar-denominated investments

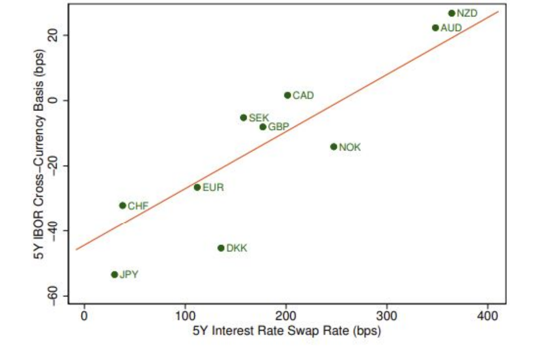
Essentially, CIP deviations can be linked to demand for US treasury bonds and by proxy USD funding, which makes the topic especially relevant to any analysis of the treasury as a safe haven

Explaining Covered Interest Parity Deviations

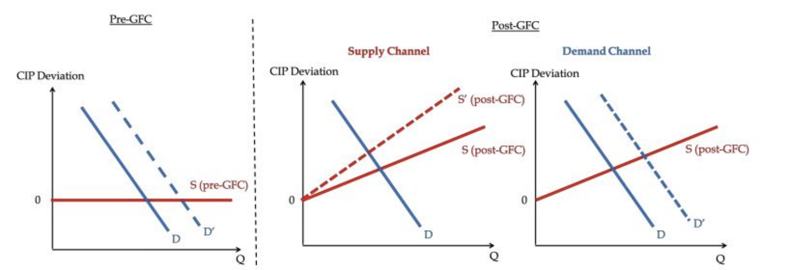
Examining the Math behind CIP deviations:

- CIP deviations, or the cross-currency basis, are traditionally measured by this equation: $x^{i/\$}_{t,t+n} = \underbrace{y^{\$}_{t,t+n}}_{\text{Cash Market Dollar Rate}} - \underbrace{y^{i}_{t,t+n} - \rho^{i/\$}_{t,t+n}}_{\text{FX Swap Market Dollar Rate}}$
    - $x^{i/\$}_{t,t+n}$ is the deviation
    - $y^{\$}_{t,t+n}$ is the cash dollar rate (logarithmic)
    - $y^{i}_{t,t+n}$ is the foreign currency rate (logarithmic)
    - $\rho^{i/\$}_{t,t+n} = [\log(F^{i/\$}_{t,t+n}) - \log(S^{i/\$}_{t})]/n$ represents the differential between the Forward and the Swap
- When (LIBOR) CIP deviations are negative, as they traditionally have been post-2008, the cost of borrowing dollars in the swap market is higher than in the cash market
- However, when we measure CIP deviations on government bonds (something conceptually more complex than risk-free LIBOR), our basis is then academically defined as $\Phi^{i/\$}_{t,t+n} \equiv -x^{i/\$,Govt}_{t,t+n} = \underbrace{y^{i,Govt}_{t,t+n} - \rho^{i/\$}_{t,t+n}}_{\text{Foreign Synthetic Dollar Yield}} - \underbrace{y^{\$,Govt}_{t,t+n}}_{\text{U.S. Treasury Yield}}$

Correlation between the basis and interest rates:



Visualizing offshore dollar funding supply and demand:



Time Series

Average CIP Deviations

- Measured in bps
- I worked with deviations on the 3M, 5Y, and 10Y issuances
- Simple average of deviations of all G10 currencies
- Positive vs. Negative
  - If the basis is positive, then borrowing in the swap market is more expensive
- We measured CIP deviations in terms of government bond yields
  - Unlike LIBOR CIP deviations, these deviations aren't purely arbitrage
  - They reflect aspects like flight-to-safety

Broad Dollar Index (FED)

- How strong is the dollar compared to the currencies of all the countries the U.S. trades with?
- Weighted average based on each country's trade share.
- ~26 currencies

Investment-Grade Credit Spreads

- Measures yield-differential between BBB bonds and 10Y treasuries
- It's a gauge of investors' confidence in the market
  - Low spreads indicate optimism
  - High spreads reflect pessimism

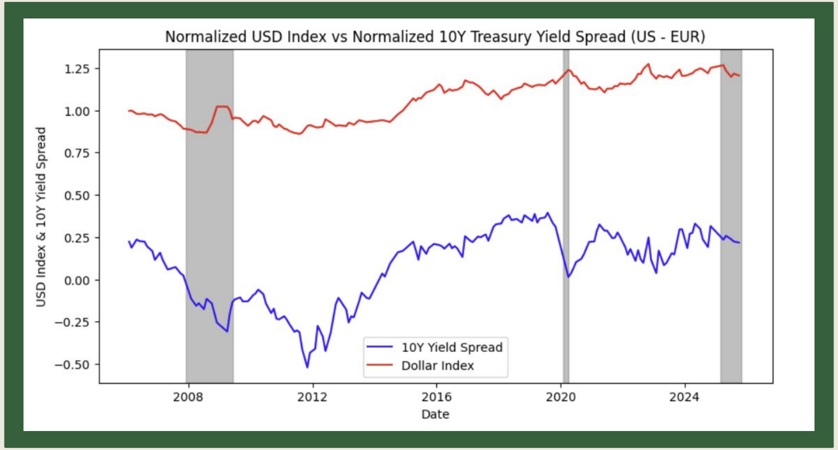
VIX

- Real-time measure of the expected volatility of the U.S. stock market over the next 30 days
- Market's fear gauge
  - Spikes: investors are anxious; markets fall

US-EU Yield Spread (US10Y - EU10Y)

- Measures relative yield US treasuries offer relative to Eurozone benchmarks
- A positive spread indicates higher relative US yields
- Useful for further examining flight-to-safety mechanisms

The Relationship Between Rate Spreads and the Dollar



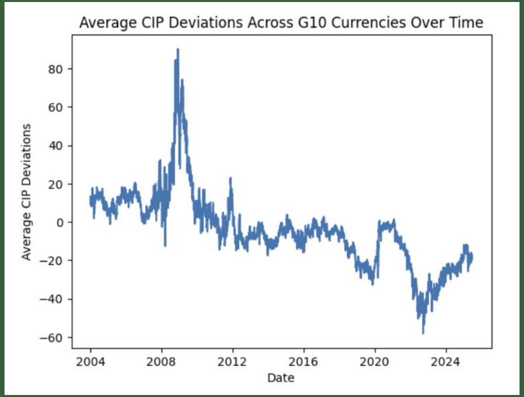
Notes: To examine the 'specialness' of US government bonds, I've looked at historical data mapping the differential between US and Eurozone 10Y Yields with the USD Broad Index

Between the two recessionary periods in 2009 and 2019, the correlation between the Yield Differential and the USD was 0.887 which is quite high. During the two recessionary periods looked at, the dollar index appreciated despite a weakening of the US 10Y Yield, demonstrating the 'specialness' of the dollar and flight to safety

Interestingly, when we look at the data post-Liberation Day, we see a switch with a weakening of the USD despite a jump in yields in April. For the year of 2025, despite not being in a recession, relative yield strength has only had a 0.332 correlation with the USD, demonstrating a somewhat perplexing anomaly

Closer Look at CIP Deviations on the 5 Year:

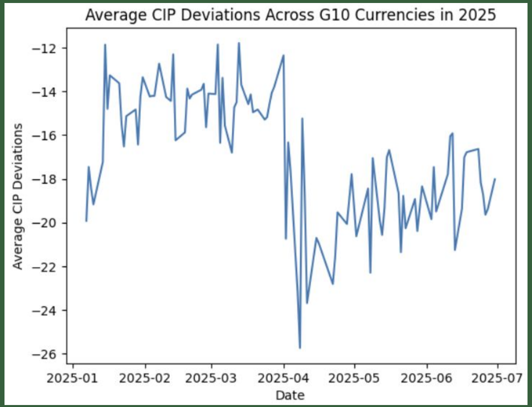
Historical Data



Notes: In the wake of the 2008 Financial Crisis, the Cross-Currency Basis became largely negative, demonstrating an erosion of long-term government yield 'specialness'

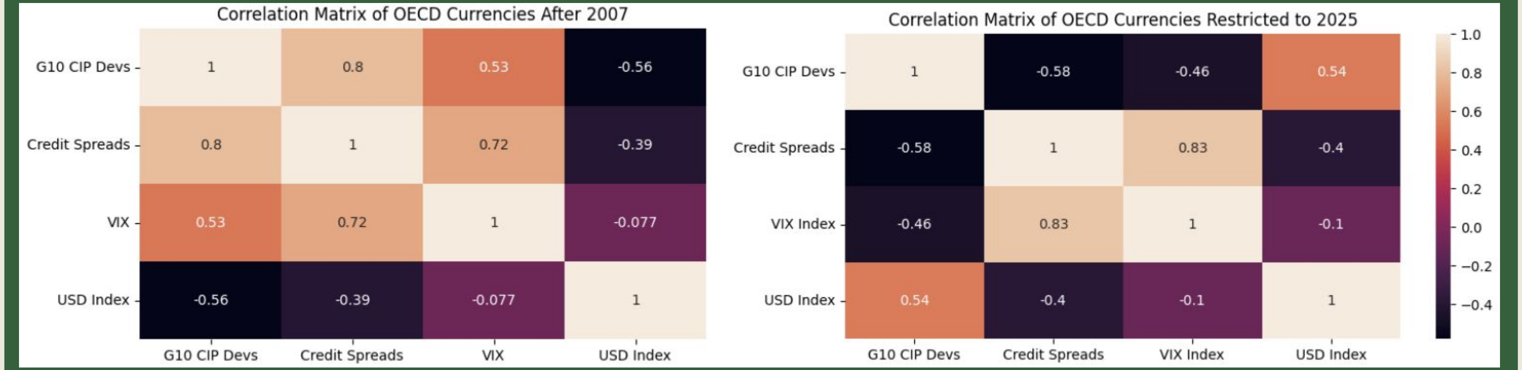
During the two recessions, the basis saw sharp upward swings, demonstrating flight-to-safety

Post-Liberation Day



Notes: The amount of data on hand is insufficient to establish conclusive a conclusive trend. What is evident is that a significant degree of volatility occurred in the cross-currency basis around April, signifying that the large degree of capital flight from treasuries had an impact

Examining Correlation Heatmaps of CIP Deviations



Notes: Using the 5Y CIP deviation time series, we map its correlation with Investment-Grade Credit Spreads, the VIX Index, and the Broad US Dollar Index. We see that historically government bond CIP Deviations have positive correlations with Credit Spreads and the VIX

If the market became more volatile, as measured by VIX or Credit Spreads, the cost of borrowing 5 Year bonds in the swap market would increase due to the flight to safety mechanism. In periods when the dollar was weak, presumably due to a more stable world economy, the dollar's 'specialness' would diminish

Interestingly, when we restrict our data to 2025; we see that all of the factors' correlations with CIP Deviations reverse in the post-Liberation Day Environment

Mapping CIP Deviations With a Regression

Historical Post-GFC Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            G10 CIP Devs   R-squared:                       0.722
Model:                             OLS   Adj. R-squared:                  0.722
Method:                  Least Squares   F-statistic:                     3347.
Date:                 Fri, 12 Dec 2025   Prob (F-statistic):               0.00
Time:                         02:41:00   Log-Likelihood:                -14655.
No. Observations:                 3874   AIC:                         2.932e+04
Df Residuals:                     3870   BIC:                         2.934e+04
Df Model:                            3
Covariance Type:             nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const             15.1547      1.783      8.500      0.000      11.659      18.650
Credit Spreads    12.0605      0.241     49.997      0.000      11.588      12.533
VIX               -0.0033      0.028     -0.117      0.907      -0.059       0.052
USD Index         -0.4435      0.016    -28.213      0.000      -0.474      -0.413
==============================================================================
```

Notes: For post-2007 data , a multilinear regression mapping CIP deviations on the 3 factors yields:

CIP Devs = 15.15 + 12.06(Credit Spreads) - 0.44(USD Index)

I rejected the VIX in my regression because it's statistically insignificant according to the data

Importantly, 72.2% of the data is explained by the factors, indicating the regression's relative strength

Post-Liberation Day Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            G10 CIP Devs   R-squared:                       0.485
Model:                             OLS   Adj. R-squared:            [illegible]
Method:                  Least Squares   F-statistic:                     29.81
Date:                 Fri, 12 Dec 2025   Prob (F-statistic):           1.14e-13
Time:                         02:39:02   Log-Likelihood:                -215.49
No. Observations:                   99   AIC:                             439.0
Df Residuals:                       95   BIC:                             449.4
Df Model:                            3
Covariance Type:             nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const            -68.0401     13.234     -5.141      0.000     -94.312     -41.768
Credit Spreads    -4.1285      3.737     -1.105      0.272     -11.547       3.291
VIX               -0.1221      0.066     -1.839      0.069      -0.254       0.010
USD Index          0.4664      0.090      5.197      0.000       0.288       0.644
==============================================================================
```

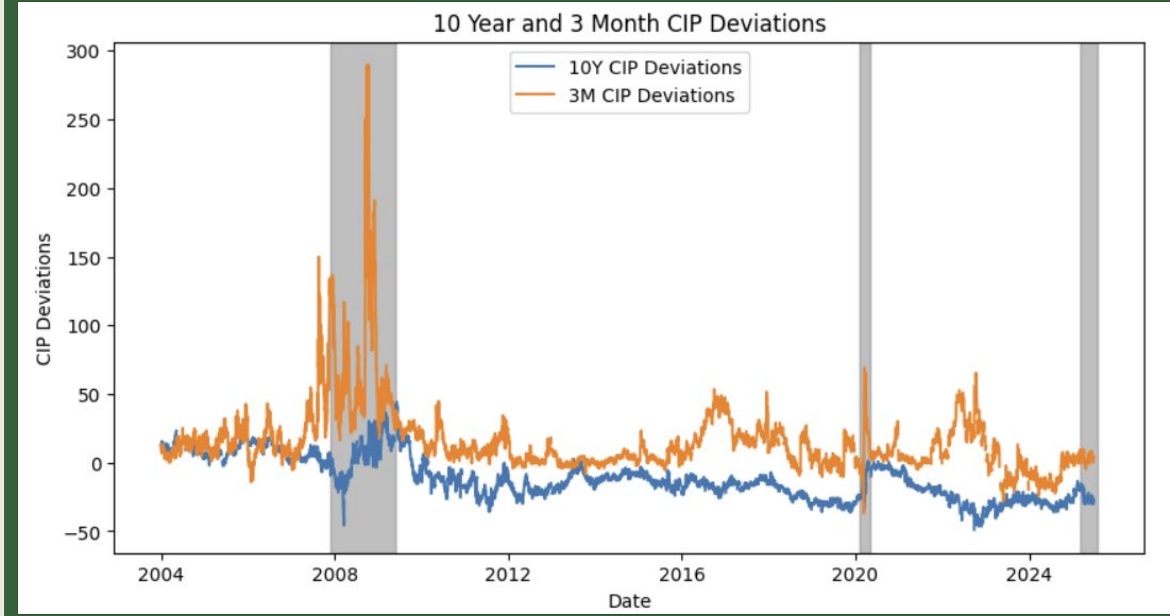
Notes: In the post-Liberation Day environment, our new regression yields:

CIP Devs = -68.04 - 4.13(Credit Spreads) - 0.12 (VIX) + 0.44(USD Index)

Credit Spreads diminish in statistical significance while the VIX becomes a more powerful indicator. The coefficients also switch their signs

Only 48.5% of the data is explained by the factors, indicating this regression is much less powerful

Analyzing CIP Deviations in the 3 Month and the 10Y



Notes: While the 3 Month and the 10 Year tracked each other closely before the GFC, the two series consistently diverged following the crisis

The 3 Month CIP Deviation on Government Bonds has remained consistently positive, and would see much starker increases during recessions, than the 10 Year

This shows that 3 Month Bond's status as a flight-to-safety mechanism continues to remain 'special'

The weakness of the US 10 Year Cash Rate demonstrates investors' increased concerns about the long-term viability of US government offerings

Additionally in Liberation Day, the 10Y saw significantly more sustained depression in the cash rate compared to the 3 Month, demonstrating the loss of the 10Y's 'specialness'